Expert Conjectures

Abducer

KB ref: p-vs-np
Executed discovery branches: 25
Expert conjecture packages: 90
Ready for prove: 90
Ready for discovery: 0
Restricted-regime programmes: 7

Judgment

- Discovery now compiles branch executions into theorem packages rather than leaving them as abstract branch prompts.
- Next move: Promote the ready-for-prove expert conjectures into the proof queue and keep the restricted-regime and foundation-search programmes on the discovery frontier.

Ready For Prove

- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.10: There exists an admissible 'MCSP obstruction family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.09: No admissible 'MCSP obstruction family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.
- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.08: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'There exist

explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.07: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' under the current route commitments.

- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.06: For every admissible 'MCSP witness family', there is a transport law from 'nonuniform-circuits' to 'formula-size' that preserves 'MCSP complexity profile in nonuniform-circuits' and retains enough strength to activate 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.05: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.04: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.

- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.03: There exists an admissible 'MCSP obstruction family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.02: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial

lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.01: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-pattern-regime-localization.10: There exists an admissible 'MCSP obstruction family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-pattern-regime-localization.09: No admissible 'MCSP obstruction family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.

- conjpkg.disclayer-p-vs-np-pattern-regime-localization.08: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- conjpkg.disclayer-p-vs-np-pattern-regime-localization.07: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'There exist explicit"meta"problems (including MCSP variants) such that proving certain

emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' under the current route commitments.

- conjpkg.disclayer-p-vs-np-pattern-regime-localization.06: For every admissible 'MCSP witness family', there is a transport law from 'nonuniform-circuits' to 'formula-size' that preserves 'MCSP complexity profile in nonuniform-circuits' and retains enough strength to activate 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- conjpkg.disclayer-p-vs-np-pattern-regime-localization.05: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-pattern-regime-localization.04: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.

- conjpkg.disclayer-p-vs-np-pattern-regime-localization.03: There exists an admissible 'MCSP obstruction family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-pattern-regime-localization.02: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- conjpkg.disclayer-p-vs-np-pattern-regime-localization.01: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.13: There exists an intermediate typed object built from 'MCSP obstruction family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through theorem-sized maps preserving 'MCSP complexity profile in nonuniform-circuits'.

- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.12: There exists an admissible 'MCSP obstruction family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.11: No admissible 'MCSP obstruction family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.

- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.10: There exists an intermediate typed object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' factors through theorem-sized maps preserving 'MCSP complexity profile in nonuniform-circuits'.

- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.09: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.08: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' under the current route commitments.

- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.07: For every admissible 'MCSP witness family', there is a transport law from 'nonuniform-circuits' to 'formula-size' that preserves 'MCSP complexity profile in nonuniform-circuits' and retains enough strength to activate 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.06: There exists an intermediate typed object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through theorem-sized maps preserving 'MCSP complexity profile in nonuniform-circuits'.

- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.05: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.04: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits'

and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.

- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.03: There exists an admissible 'MCSP obstruction family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.02: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.01: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-family-abstraction-search.10: There exists an admissible 'MCSP obstruction family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-family-abstraction-search.09: No admissible 'MCSP obstruction family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^

1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.

- conjpkg.disclayer-p-vs-np-family-abstraction-search.08: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- conjpkg.disclayer-p-vs-np-family-abstraction-search.07: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' under the current route commitments.

- conjpkg.disclayer-p-vs-np-family-abstraction-search.06: For every admissible 'MCSP witness family', there is a transport law from 'nonuniform-circuits' to 'formula-size' that preserves 'MCSP complexity profile in nonuniform-circuits' and retains enough strength to activate 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

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- conjpkg.disclayer-p-vs-np-family-abstraction-search.01: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.06: There exists an admissible 'unspecified reduction gadget family' whose realization internalizes 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.' while preserving 'consequence schema NP-complete-classification' strongly enough to support 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.'.

- conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.05: No admissible 'unspecified reduction gadget family' can simultaneously preserve 'consequence schema NP-complete-classification' and realize 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' under the current route commitments.

- conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.04: There exists an admissible 'unspecified witness family' whose realization internalizes 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.' while preserving 'consequence schema NP-complete-classification' strongly enough to support 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops

evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.'.

- conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.03: No admissible 'unspecified witness family' can simultaneously preserve 'consequence schema NP-complete-classification' and realize 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' under the current route commitments.

- conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.02: There exists an admissible 'unspecified reduction gadget family' such that 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.' can be organized into a route preserving 'consequence schema NP-complete-classification' and thereby supports 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.'.

- conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.01: There exists an admissible 'unspecified witness family' such that 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.' can be organized into a route preserving 'consequence schema NP-complete-classification' and thereby supports 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.'.

- conjpkg.disclayer-p-vs-np-family-explicitness.08: There exists an admissible 'PIT witness family' whose realization internalizes 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' while preserving 'PIT semantic object' strongly enough to support 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.

- conjpkg.disclayer-p-vs-np-family-explicitness.07: No admissible 'PIT witness family' can simultaneously preserve 'PIT semantic object' and realize 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.' under the current route commitments.

- conjpkg.disclayer-p-vs-np-family-explicitness.06: For every admissible 'PIT witness family', there is a transport law from 'black-box-PIT' to 'arithmetic-circuits' that preserves 'PIT semantic object' and retains enough strength to activate 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.

- conjpkg.disclayer-p-vs-np-family-explicitness.05: There exists an admissible 'PIT certificate family' whose realization internalizes 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' while preserving 'PIT semantic object' strongly enough to support 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.

- conjpkg.disclayer-p-vs-np-family-explicitness.04: No admissible 'PIT certificate family' can simultaneously preserve 'PIT semantic object' and realize 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.' under the current route commitments.

- conjpkg.disclayer-p-vs-np-family-explicitness.03: For every admissible 'PIT certificate family', there is a transport law from 'black-box-PIT' to 'arithmetic-circuits' that preserves 'PIT semantic object' and retains enough strength to activate 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.

- conjpkg.disclayer-p-vs-np-family-explicitness.02: There exists an admissible 'PIT witness family' such that 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' can be organized into a route preserving 'PIT semantic object' and thereby supports 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.

- conjpkg.disclayer-p-vs-np-family-explicitness.01: There exists an admissible 'PIT certificate family' such that 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' can be organized into a route preserving 'PIT semantic object' and thereby supports 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.13: There exists an admissible 'MCSP obstruction family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.12: There exists an intermediate typed object built from 'MCSP obstruction family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through theorem-sized maps preserving 'MCSP complexity profile in nonuniform-circuits'.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.11: No admissible 'MCSP obstruction family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.10: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.09: There exists an intermediate typed object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' factors through theorem-sized maps preserving 'MCSP complexity profile in nonuniform-circuits'.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.08: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' under the current route commitments.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.07: For every admissible 'MCSP witness family', there is a transport law from 'nonuniform-circuits' to 'formula-size' that preserves 'MCSP complexity profile in nonuniform-circuits' and retains enough strength to ac-

tivate 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.06: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.05: There exists an intermediate typed object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through theorem-sized maps preserving 'MCSP complexity profile in nonuniform-circuits'.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.04: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.03: There exists a mediating object built from 'MCSP obstruction family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.03: There exists a mediating object built from 'MCSP obstruction family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n

) imply major consequences such as NP not in P/poly.' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.02: There exists a mediating object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.02: There exists a mediating object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.01: There exists a mediating object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute.

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.01: There exists a mediating object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.14: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'exponential-lower-bound witness complexity' strongly enough to support 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.13: No admissible 'MCSP witness family' can simultaneously preserve 'exponential-lower-bound witness complexity' and realize 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.' under the current route commitments.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.12: For every admissible 'MCSP witness family', the route-side and target-side operationalizations are equivalent in the sense needed to transport 'exponential-lower-bound witness complexity' toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.11: For every admissible 'MCSP witness family', there is a transport law from 'nonuniform-circuits' to 'unspecified' that preserves 'exponential-lower-bound witness complexity' and retains enough strength to activate 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.10: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.09: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial

formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' under the current route commitments.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.08: For every admissible 'MCSP witness family', the route-side and target-side operationalizations are equivalent in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.07: For every admissible 'MCSP witness family', there is a transport law from 'nonuniform-circuits' to 'formula-size' that preserves 'MCSP complexity profile in nonuniform-circuits' and retains enough strength to activate 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.06: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.05: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.04: For every admissible 'MCSP witness family', the route-side and target-side operationalizations are equivalent in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.03: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known

lower bounds in some settings.', and 'exponential-lower-bound witness complexity' transports coherently along that witness.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.03: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.', and 'exponential-lower-bound witness complexity' transports coherently along that witness.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.02: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).', and 'MCSP complexity profile in nonuniform-circuits' transports coherently along that witness.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.02: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).', and 'MCSP complexity profile in nonuniform-circuits' transports coherently along that witness.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.01: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.', and 'MCSP complexity profile in nonuniform-circuits' transports coherently along that witness.

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.01: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.', and 'MCSP complexity profile in nonuniform-circuits' transports coherently along that witness.

Restricted-Regime Programmes

- Special-Case Theorem: invariant-discovery: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content

while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.

- Threshold Version: regime-localization: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.
- Special-Case Theorem: hybridization-of-partial-bridges: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.
- Special-Case Theorem: route-as-commutative-diagram: There exists a mediating object M and typed morphisms making the route commute; otherwise the failure is certified as no morphism, non-functorial transport, broken naturality, or missing universal property.
- Special-Case Theorem: route-as-commutative-diagram: There exists a mediating object M and typed morphisms making the route commute; otherwise the failure is certified as no morphism, non-functorial transport, broken naturality, or missing universal property.
- Equivalence Theorem: route-as-equivalence-problem: Either exhibit an equivalence E between the live formulations and prove that the target-relevant invariant transports coherently along E, or prove that no such equivalence witness exists in the current regime.
- Equivalence Theorem: route-as-equivalence-problem: Either exhibit an equivalence E between the live formulations and prove that the target-relevant invariant transports coherently along E, or prove that no such equivalence witness exists in the current regime.

Foundation-Search Programmes

- conjpkg.disclayer-p-vs-np-pattern-regime-localization.10: threshold and regime-restriction frameworks, restricted-model lower-bound programmes, proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-pattern-regime-localization.09: threshold and regime-restriction frameworks, restricted-model lower-bound programmes, proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-pattern-regime-localization.08: threshold and regime-restriction frameworks, restricted-model lower-bound programmes, proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-pattern-regime-localization.07: threshold and regime-restriction frameworks, restricted-model lower-bound programmes, proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates

- conjpkg.disclayer-p-vs-np-pattern-regime-localization.06: threshold and regime-restriction frameworks, restricted-model lower-bound programmes, proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-pattern-regime-localization.05: threshold and regime-restriction frameworks, restricted-model lower-bound programmes, proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-pattern-regime-localization.04: threshold and regime-restriction frameworks, restricted-model lower-bound programmes, proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-pattern-regime-localization.03: threshold and regime-restriction frameworks, restricted-model lower-bound programmes, proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-pattern-regime-localization.02: threshold and regime-restriction frameworks, restricted-model lower-bound programmes, proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-pattern-regime-localization.01: threshold and regime-restriction frameworks, restricted-model lower-bound programmes, proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.10: potential / energy method frameworks, monotone measure frameworks, certificate-complexity and witness-measure theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.09: potential / energy method frameworks, monotone measure frameworks, certificate-complexity and witness-measure theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.08: potential / energy method frameworks, monotone measure frameworks, certificate-complexity and witness-measure theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.07: potential / energy method frameworks, monotone measure frameworks, certificate-complexity and witness-measure theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.06: potential / energy method frameworks, monotone measure frameworks, certificate-complexity and witness-measure theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.05: potential / energy method frameworks, monotone measure frameworks, certificate-complexity and witness-measure theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.04: potential / energy method frameworks, monotone measure frameworks, certificate-complexity and witness-measure theories, proof complexity, pseudorandomness / derandomization

- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.03: potential / energy method frameworks, monotone measure frameworks, certificate-complexity and witness-measure theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.02: potential / energy method frameworks, monotone measure frameworks, certificate-complexity and witness-measure theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.01: potential / energy method frameworks, monotone measure frameworks, certificate-complexity and witness-measure theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.13: composed transfer frameworks, hybrid argument methods, multi-interface reduction theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.12: composed transfer frameworks, hybrid argument methods, multi-interface reduction theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.11: composed transfer frameworks, hybrid argument methods, multi-interface reduction theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.10: composed transfer frameworks, hybrid argument methods, multi-interface reduction theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.09: composed transfer frameworks, hybrid argument methods, multi-interface reduction theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.08: composed transfer frameworks, hybrid argument methods, multi-interface reduction theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.07: composed transfer frameworks, hybrid argument methods, multi-interface reduction theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.06: composed transfer frameworks, hybrid argument methods, multi-interface reduction theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.05: composed transfer frameworks, hybrid argument methods, multi-interface reduction theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.04: composed transfer frameworks, hybrid argument methods, multi-interface reduction theories, proof complexity, pseudorandomness / derandomization

- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.03: composed transfer frameworks, hybrid argument methods, multi-interface reduction theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.02: composed transfer frameworks, hybrid argument methods, multi-interface reduction theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.01: composed transfer frameworks, hybrid argument methods, multi-interface reduction theories, proof complexity, pseudorandomness / derandomization
- conjpkg.disclayer-p-vs-np-family-explicitness.08: derandomization and generator frameworks, algebraic certificate constructions, extractor / hitter frameworks
- conjpkg.disclayer-p-vs-np-family-explicitness.07: derandomization and generator frameworks, algebraic certificate constructions, extractor / hitter frameworks
- conjpkg.disclayer-p-vs-np-family-explicitness.06: derandomization and generator frameworks, algebraic certificate constructions, extractor / hitter frameworks
- conjpkg.disclayer-p-vs-np-family-explicitness.05: derandomization and generator frameworks, algebraic certificate constructions, extractor / hitter frameworks
- conjpkg.disclayer-p-vs-np-family-explicitness.04: derandomization and generator frameworks, algebraic certificate constructions, extractor / hitter frameworks
- conjpkg.disclayer-p-vs-np-family-explicitness.03: derandomization and generator frameworks, algebraic certificate constructions, extractor / hitter frameworks
- conjpkg.disclayer-p-vs-np-family-explicitness.02: derandomization and generator frameworks, algebraic certificate constructions, extractor / hitter frameworks
- conjpkg.disclayer-p-vs-np-family-explicitness.01: derandomization and generator frameworks, algebraic certificate constructions, extractor / hitter frameworks
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.14: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.13: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.12: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.11: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions

- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.10: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.09: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.08: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.07: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.06: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.05: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.04: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.03: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.03: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.02: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.02: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.01: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions
- conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.01: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.13: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory
- conjpkg.disclayer-p-vs-np-family-categorical-reframing.12: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory
- conjpkg.disclayer-p-vs-np-family-categorical-reframing.11: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory
- conjpkg.disclayer-p-vs-np-family-categorical-reframing.10: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory
- conjpkg.disclayer-p-vs-np-family-categorical-reframing.09: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory
- conjpkg.disclayer-p-vs-np-family-categorical-reframing.08: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory
- conjpkg.disclayer-p-vs-np-family-categorical-reframing.07: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory
- conjpkg.disclayer-p-vs-np-family-categorical-reframing.06: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory
- conjpkg.disclayer-p-vs-np-family-categorical-reframing.05: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory
- conjpkg.disclayer-p-vs-np-family-categorical-reframing.04: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory
- conjpkg.disclayer-p-vs-np-family-categorical-reframing.03: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory
- conjpkg.disclayer-p-vs-np-family-categorical-reframing.03: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory
- conjpkg.disclayer-p-vs-np-family-categorical-reframing.02: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory

- conjpkg.disclayer-p-vs-np-family-categorical-reframing.02: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory
- conjpkg.disclayer-p-vs-np-family-categorical-reframing.01: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory
- conjpkg.disclayer-p-vs-np-family-categorical-reframing.01: sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory
- conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.06: lower-bound barrier frameworks, local-reduction impossibility theory, gadget and locality-preservation frameworks
- conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.05: lower-bound barrier frameworks, local-reduction impossibility theory, gadget and locality-preservation frameworks
- conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.04: lower-bound barrier frameworks, local-reduction impossibility theory, gadget and locality-preservation frameworks
- conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.03: lower-bound barrier frameworks, local-reduction impossibility theory, gadget and locality-preservation frameworks
- conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.02: lower-bound barrier frameworks, local-reduction impossibility theory, gadget and locality-preservation frameworks
- conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.01: lower-bound barrier frameworks, local-reduction impossibility theory, gadget and locality-preservation frameworks
- conjpkg.disclayer-p-vs-np-family-abstraction-search.10: proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-family-abstraction-search.09: proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-family-abstraction-search.08: proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-family-abstraction-search.07: proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-family-abstraction-search.06: proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-family-abstraction-search.05: proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-family-abstraction-search.04: proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-family-abstraction-search.03: proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates

- conjpkg.disclayer-p-vs-np-family-abstraction-search.02: proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates
- conjpkg.disclayer-p-vs-np-family-abstraction-search.01: proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates

Packages

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.01

- Exact statement: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.', and 'MCSP complexity profile in nonuniform-circuits' transports coherently along that witness.
- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' is not licensed. no_equivalence_witness

- Proof strategy family: equivalence-transport
- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.
- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'MCSP complexity profile in nonuniform-circuits' to the target-side operationalization. Local negative probe: no_equivalence_witness.
- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.
- Candidate lemmas: Equivalence and coherent transport theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family
- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.02

- Exact statement: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).', and 'MCSP complexity profile in nonuniform-circuits' transports coherently along that witness.
- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.

- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' is not licensed. no_equivalence_witness
- Proof strategy family: equivalence-transport
- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.
- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'MCSP complexity profile in nonuniform-circuits' to the target-side operationalization. Local negative probe: no_equivalence_witness.
- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.
- Candidate lemmas: Equivalence and coherent transport theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family
- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.03

- Exact statement: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.', and 'exponential-lower-bound witness complexity' transports coherently along that witness.
- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'exponential-lower-bound witness complexity'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'exponential-lower-bound witness complexity' toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.atserias-m-uller-simple-general-magnification-fr'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.' is not licensed. no_equivalence_witness
- Proof strategy family: equivalence-transport
- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.
- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'exponential-lower-bound witness complexity' to the target-side operationalization. Local negative probe: no_equivalence_witness.

- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'exponential-lower-bound witness complexity' toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.
- Candidate lemmas: Equivalence and coherent transport theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family
- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.04

- Exact statement: For every admissible 'MCSP witness family', the route-side and target-side operationalizations are equivalent in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' is not licensed. no_equivalence_witness
- Proof strategy family: equivalence-transport
- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.
- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'MCSP complexity profile in nonuniform-circuits' to the target-side operationalization. Local negative probe: no_equivalence_witness.
- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.
- Candidate lemmas: Equivalence theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family
- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.05

- Exact statement: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.
- Restricted-regime mode: equivalence-theorem

- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' is not licensed. no_equivalence_witness
- Proof strategy family: equivalence-transport
- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.
- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'MCSP complexity profile in nonuniform-circuits' to the target-side operationalization. Local negative probe: no_equivalence_witness.
- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.

- Candidate lemmas: Obstruction theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family
- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.06

- Exact statement: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and

input length N=2^n) imply major consequences such as NP not in P/poly.' is not licensed. no_equivalence_witness

- Proof strategy family: equivalence-transport
- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.
- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'MCSP complexity profile in nonuniform-circuits' to the target-side operationalization. Local negative probe: no_equivalence_witness.
- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.
- Candidate lemmas: Realization theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family
- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.07

- Exact statement: For every admissible 'MCSP witness family', there is a transport law from 'nonuniform-circuits' to 'formula-size' that preserves 'MCSP complexity profile in nonuniform-circuits' and retains enough strength to activate 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It

must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.

- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly)'.

- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.

- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).

- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly)' is not licensed. no_equivalence_witness

- Proof strategy family: equivalence-transport

- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.

- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'MCSP complexity profile in nonuniform-circuits' to the target-side operationalization. Local negative probe: no_equivalence_witness.

- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly)'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.

- Candidate lemmas: Transport theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family

- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.08

- Exact statement: For every admissible 'MCSP witness family', the route-side and target-side operationalizations are equivalent in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' is not licensed. no_equivalence_witness

- Proof strategy family: equivalence-transport
- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.
- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'MCSP complexity profile in nonuniform-circuits' to the target-side operationalization. Local negative probe: no_equivalence_witness.
- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.
- Candidate lemmas: Equivalence theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family
- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.09

- Exact statement: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' under the current route commitments.
- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It

must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.

- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly)'.

- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.

- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).

- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly)' is not licensed. no_equivalence_witness

- Proof strategy family: equivalence-transport

- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.

- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'MCSP complexity profile in nonuniform-circuits' to the target-side operationalization. Local negative probe: no_equivalence_witness.

- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly)'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.

- Candidate lemmas: Obstruction theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family

- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.10

- Exact statement: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g.

formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' is not licensed. no_equivalence_witness

- Proof strategy family: equivalence-transport
- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.
- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'MCSP complexity profile in nonuniform-circuits' to the target-side operationalization. Local negative probe: no_equivalence_witness.
- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.
- Candidate lemmas: Realization theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family
- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.11

- Exact statement: For every admissible 'MCSP witness family', there is a transport law from 'nonuniform-circuits' to 'unspecified' that preserves 'exponential-lower-bound witness complexity' and retains enough strength to activate 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.
- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model

implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'exponential-lower-bound witness complexity'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.

- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'exponential-lower-bound witness complexity' toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.atserias-m-uller-simple-general-magnification-fr'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.' is not licensed. no_equivalence_witness
- Proof strategy family: equivalence-transport
- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.
- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'exponential-lower-bound witness complexity' to the target-side operationalization. Local negative probe: no_equivalence_witness.
- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'exponential-lower-bound witness complexity' toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.
- Candidate lemmas: Transport theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family

- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.12

- Exact statement: For every admissible 'MCSP witness family', the route-side and target-side operationalizations are equivalent in the sense needed to transport 'exponential-lower-bound witness complexity' toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.
- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'exponential-lower-bound witness complexity'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'exponential-lower-bound witness complexity' toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.atserias-m-uller-simple-general-magnification-fr'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.' is not licensed. no_equivalence_witness
- Proof strategy family: equivalence-transport

- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.
- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'exponential-lower-bound witness complexity' to the target-side operationalization. Local negative probe: no_equivalence_witness.
- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'exponential-lower-bound witness complexity' toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.
- Candidate lemmas: Equivalence theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family
- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.13

- Exact statement: No admissible 'MCSP witness family' can simultaneously preserve 'exponential-lower-bound witness complexity' and realize 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.' under the current route commitments.
- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'exponential-lower-bound witness complexity'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.

- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'exponential-lower-bound witness complexity' toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.atserias-m-uller-simple-general-magnification-fr'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.' is not licensed. no_equivalence_witness
- Proof strategy family: equivalence-transport
- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.
- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'exponential-lower-bound witness complexity' to the target-side operationalization. Local negative probe: no_equivalence_witness.
- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'exponential-lower-bound witness complexity' toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.
- Candidate lemmas: Obstruction theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family
- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.14

- Exact statement: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'exponential-lower-bound witness complexity' strongly enough to support 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.
- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'exponential-lower-bound witness complexity'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'exponential-lower-bound witness complexity' toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.atserias-m-uller-simple-general-magnification-fr'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.' is not licensed. no_equivalence_witness
- Proof strategy family: equivalence-transport
- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.

- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'exponential-lower-bound witness complexity' to the target-side operationalization. Local negative probe: no_equivalence_witness.
- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'exponential-lower-bound witness complexity' toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.
- Candidate lemmas: Realization theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family
- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-categorical-reframing.01

- Exact statement: There exists a mediating object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.

- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'. non_commuting_route
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.
- Candidate lemmas: Commutative factorization theorem for MCSP witness family; Coherence diagram lemma for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-categorical-reframing.02

- Exact statement: There exists a mediating object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much

stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'. non_commuting_route

- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.
- Candidate lemmas: Commutative factorization theorem for MCSP witness family; Coherence diagram lemma for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-categorical-reframing.03

- Exact statement: There exists a mediating object built from 'MCSP obstruction family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute.
- Restricted-regime mode: special-case
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)

- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.
- First nontrivial lemma: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'. non_commuting_route
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.
- Proof obligations: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.

- Candidate lemmas: Commutative factorization theorem for MCSP obstruction family; Coherence diagram lemma for MCSP obstruction family
- Independent value: A concrete special-case theorem for 'MCSP obstruction family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-categorical-reframing.04

- Exact statement: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'. non_commuting_route

- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.
- Candidate lemmas: Obstruction theorem for MCSP witness family; Coherence diagram lemma for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-categorical-reframing.05

- Exact statement: There exists an intermediate typed object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through theorem-sized maps preserving 'MCSP complexity profile in nonuniform-circuits'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational

role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.

- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'. non_commuting_route
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.
- Candidate lemmas: Factorization theorem for MCSP witness family; Coherence diagram lemma for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

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- Exact statement: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'. non_commuting_route
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.

- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.
- Candidate lemmas: Realization theorem for MCSP witness family; Coherence diagram lemma for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-categorical-reframing.07

- Exact statement: For every admissible 'MCSP witness family', there is a transport law from 'nonuniform-circuits' to 'formula-size' that preserves 'MCSP complexity profile in nonuniform-circuits' and retains enough strength to activate 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a

strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'. non_commuting_route
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.
- Candidate lemmas: Transport theorem for MCSP witness family; Coherence diagram lemma for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-categorical-reframing.08

- Exact statement: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' under the current route commitments.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'. non_commuting_route
- Proof strategy family: constructive-realization

- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.
- Candidate lemmas: Obstruction theorem for MCSP witness family; Coherence diagram lemma for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-categorical-reframing.09

- Exact statement: There exists an intermediate typed object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' factors through theorem-sized maps preserving 'MCSP complexity profile in nonuniform-circuits'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It

must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.

- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly)'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly)'. non_commuting_route
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly)'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.
- Candidate lemmas: Factorization theorem for MCSP witness family; Coherence diagram lemma for MCSP witness family

- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-categorical-reframing.10

- Exact statement: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).

- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'. non_commuting_route
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.
- Candidate lemmas: Realization theorem for MCSP witness family; Coherence diagram lemma for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-categorical-reframing.11

- Exact statement: No admissible 'MCSP obstruction family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.
- Restricted-regime mode: special-case
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)

- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.
- First nontrivial lemma: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'. non_commuting_route
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.
- Proof obligations: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.

- Candidate lemmas: Obstruction theorem for MCSP obstruction family; Coherence diagram lemma for MCSP obstruction family
- Independent value: A concrete special-case theorem for 'MCSP obstruction family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-categorical-reframing.12

- Exact statement: There exists an intermediate typed object built from 'MCSP obstruction family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through theorem-sized maps preserving 'MCSP complexity profile in nonuniform-circuits'.
- Restricted-regime mode: special-case
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)
- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.
- First nontrivial lemma: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).

- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'. non_commuting_route
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.
- Proof obligations: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.
- Candidate lemmas: Factorization theorem for MCSP obstruction family; Coherence diagram lemma for MCSP obstruction family
- Independent value: A concrete special-case theorem for 'MCSP obstruction family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-categorical-reframing.13

- Exact statement: There exists an admissible 'MCSP obstruction family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: special-case
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)

- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.
- First nontrivial lemma: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'. non_commuting_route
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.
- Proof obligations: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.

- Candidate lemmas: Realization theorem for MCSP obstruction family; Coherence diagram lemma for MCSP obstruction family
- Independent value: A concrete special-case theorem for 'MCSP obstruction family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-explicitness.01

- Exact statement: There exists an admissible 'PIT certificate family' such that 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' can be organized into a route preserving 'PIT semantic object' and thereby supports 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Restricted-regime mode: special-case
- Bridge object: PIT certificate family (checkable certificate map or algebraic witness relation)
- Bridge definition: 'PIT certificate family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.'. It must preserve the invariant 'PIT semantic object'. Its operational role is 'Upgrade a semantic or existential algebraic object into an explicit algorithmic derandomization premise.'.
- First nontrivial lemma: For every admissible 'PIT certificate family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Quantifier signature: For every admissible 'PIT certificate family' on the route from 'st.def.hitting-set-for-pit-black-box' to 'st.thm.kabanets-impagliazzo-pit-derandomisation-implies'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'PIT certificate family' cannot be realized with invariant 'PIT semantic object', the route cannot force 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.

- Proof strategy family: constructive-realization
- Negative route: Assume 'PIT certificate family' cannot be realized while preserving 'PIT semantic object' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'PIT certificate family' cannot be realized while preserving 'PIT semantic object'.
- Proof obligations: For every admissible 'PIT certificate family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Candidate lemmas: Prove that a hitting-set or pseudorandom-generator family preserves 'PIT semantic object' while transporting the route from the black-box-PIT regime to the arithmetic-circuits regime strongly enough to derive 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard...'.
- Independent value: A concrete special-case theorem for 'PIT certificate family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-explicitness.02

- Exact statement: There exists an admissible 'PIT witness family' such that 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' can be organized into a route preserving 'PIT semantic object' and thereby supports 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Restricted-regime mode: special-case
- Bridge object: PIT witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'PIT witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.'. It must preserve the invariant 'PIT semantic object'. Its operational role is 'Upgrade a semantic or existential algebraic object into an explicit algorithmic derandomization premise.'.

- First nontrivial lemma: For every admissible 'PIT witness family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Quantifier signature: For every admissible 'PIT witness family' on the route from 'st.def.hitting-set-for-pit-black-box' to 'st.thm.kabanets-impagliazzo-pit-derandomisation-implies'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'PIT witness family' cannot be realized with invariant 'PIT semantic object', the route cannot force 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'PIT witness family' cannot be realized while preserving 'PIT semantic object' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'PIT witness family' cannot be realized while preserving 'PIT semantic object'.
- Proof obligations: For every admissible 'PIT witness family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Candidate lemmas: Prove that a hitting-set or pseudorandom-generator family preserves 'PIT semantic object' while transporting the route from the black-box-PIT regime to the arithmetic-circuits regime strongly enough to derive 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard...'.
- Independent value: A concrete special-case theorem for 'PIT witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-explicitness.03

- Exact statement: For every admissible 'PIT certificate family', there is a transport law from 'black-box-PIT' to 'arithmetic-circuits' that preserves 'PIT semantic object' and retains enough strength to activate 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Restricted-regime mode: special-case
- Bridge object: PIT certificate family (checkable certificate map or algebraic witness relation)
- Bridge definition: 'PIT certificate family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.'. It must preserve the invariant 'PIT semantic object'. Its operational role is 'Upgrade a semantic or existential algebraic object into an explicit algorithmic derandomization premise.'.
- First nontrivial lemma: For every admissible 'PIT certificate family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Quantifier signature: For every admissible 'PIT certificate family' on the route from 'st.def.hitting-set-for-pit-black-box' to 'st.thm.kabanets-impagliazzo-pit-derandomisation-implies'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'PIT certificate family' cannot be realized with invariant 'PIT semantic object', the route cannot force 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'PIT certificate family' cannot be realized while preserving 'PIT semantic object' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'PIT certificate family' cannot be realized while preserving 'PIT semantic object'.
- Proof obligations: For every admissible 'PIT certificate family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero

polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.

- Candidate lemmas: Transport theorem for PIT certificate family
- Independent value: A concrete special-case theorem for 'PIT certificate family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-explicitness.04

- Exact statement: No admissible 'PIT certificate family' can simultaneously preserve 'PIT semantic object' and realize 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.' under the current route commitments.
- Restricted-regime mode: special-case
- Bridge object: PIT certificate family (checkable certificate map or algebraic witness relation)
- Bridge definition: 'PIT certificate family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.'. It must preserve the invariant 'PIT semantic object'. Its operational role is 'Upgrade a semantic or existential algebraic object into an explicit algorithmic derandomization premise.'.
- First nontrivial lemma: For every admissible 'PIT certificate family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Quantifier signature: For every admissible 'PIT certificate family' on the route from 'st.def.hitting-set-for-pit-black-box' to 'st.thm.kabanets-impagliazzo-pit-derandomisation-implies'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'PIT certificate family' cannot be realized with invariant 'PIT semantic object', the route cannot force 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then

at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.

- Proof strategy family: constructive-realization
- Negative route: Assume 'PIT certificate family' cannot be realized while preserving 'PIT semantic object' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'PIT certificate family' cannot be realized while preserving 'PIT semantic object'.
- Proof obligations: For every admissible 'PIT certificate family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Candidate lemmas: Obstruction theorem for PIT certificate family
- Independent value: A concrete special-case theorem for 'PIT certificate family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-explicitness.05

- Exact statement: There exists an admissible 'PIT certificate family' whose realization internalizes 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' while preserving 'PIT semantic object' strongly enough to support 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Restricted-regime mode: special-case
- Bridge object: PIT certificate family (checkable certificate map or algebraic witness relation)
- Bridge definition: 'PIT certificate family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.'. It must preserve the invariant 'PIT semantic object'. Its operational role is 'Upgrade a semantic or existential algebraic object into an explicit algorithmic derandomization premise.'.

- First nontrivial lemma: For every admissible 'PIT certificate family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Quantifier signature: For every admissible 'PIT certificate family' on the route from 'st.def.hitting-set-for-pit-black-box' to 'st.thm.kabanets-impagliazzo-pit-derandomisation-implies'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'PIT certificate family' cannot be realized with invariant 'PIT semantic object', the route cannot force 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'PIT certificate family' cannot be realized while preserving 'PIT semantic object' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'PIT certificate family' cannot be realized while preserving 'PIT semantic object'.
- Proof obligations: For every admissible 'PIT certificate family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Candidate lemmas: Realization theorem for PIT certificate family
- Independent value: A concrete special-case theorem for 'PIT certificate family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-explicitness.06

- Exact statement: For every admissible 'PIT witness family', there is a transport law from 'black-box-PIT' to 'arithmetic-circuits' that preserves 'PIT semantic object' and retains enough strength to activate 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the

following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.

- Restricted-regime mode: special-case
- Bridge object: PIT witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'PIT witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.'. It must preserve the invariant 'PIT semantic object'. Its operational role is 'Upgrade a semantic or existential algebraic object into an explicit algorithmic derandomization premise.'.
- First nontrivial lemma: For every admissible 'PIT witness family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Quantifier signature: For every admissible 'PIT witness family' on the route from 'st.def.hitting-set-for-pit-black-box' to 'st.thm.kabanets-impagliazzo-pit-derandomisation-implies'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'PIT witness family' cannot be realized with invariant 'PIT semantic object', the route cannot force 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'PIT witness family' cannot be realized while preserving 'PIT semantic object' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'PIT witness family' cannot be realized while preserving 'PIT semantic object'.
- Proof obligations: For every admissible 'PIT witness family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.

- Candidate lemmas: Transport theorem for PIT witness family
- Independent value: A concrete special-case theorem for 'PIT witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-explicitness.07

- Exact statement: No admissible 'PIT witness family' can simultaneously preserve 'PIT semantic object' and realize 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.' under the current route commitments.
- Restricted-regime mode: special-case
- Bridge object: PIT witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'PIT witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.'. It must preserve the invariant 'PIT semantic object'. Its operational role is 'Upgrade a semantic or existential algebraic object into an explicit algorithmic derandomization premise.'.
- First nontrivial lemma: For every admissible 'PIT witness family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Quantifier signature: For every admissible 'PIT witness family' on the route from 'st.def.hitting-set-for-pit-black-box' to 'st.thm.kabanets-impagliazzo-pit-derandomisation-implies'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'PIT witness family' cannot be realized with invariant 'PIT semantic object', the route cannot force 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Proof strategy family: constructive-realization

- Negative route: Assume 'PIT witness family' cannot be realized while preserving 'PIT semantic object' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'PIT witness family' cannot be realized while preserving 'PIT semantic object'.
- Proof obligations: For every admissible 'PIT witness family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Candidate lemmas: Obstruction theorem for PIT witness family
- Independent value: A concrete special-case theorem for 'PIT witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-explicitness.08

- Exact statement: There exists an admissible 'PIT witness family' whose realization internalizes 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' while preserving 'PIT semantic object' strongly enough to support 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Restricted-regime mode: special-case
- Bridge object: PIT witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'PIT witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.'. It must preserve the invariant 'PIT semantic object'. Its operational role is 'Upgrade a semantic or existential algebraic object into an explicit algorithmic derandomization premise.'.
- First nontrivial lemma: For every admissible 'PIT witness family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H

with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.

- Quantifier signature: For every admissible 'PIT witness family' on the route from 'st.def.hitting-set-for-pit-black-box' to 'st.thm.kabanets-impagliazzo-pit-derandomisation-implies'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'PIT witness family' cannot be realized with invariant 'PIT semantic object', the route cannot force 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'PIT witness family' cannot be realized while preserving 'PIT semantic object' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'PIT witness family' cannot be realized while preserving 'PIT semantic object'.
- Proof obligations: For every admissible 'PIT witness family', the first route-specific realization step preserves 'PIT semantic object' and prepares the passage from 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' to 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Candidate lemmas: Realization theorem for PIT witness family
- Independent value: A concrete special-case theorem for 'PIT witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.01

- Exact statement: There exists an admissible 'unspecified witness family' such that 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.' can be organized into a route preserving 'consequence schema NP-complete-classification' and thereby supports 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.'.

- Restricted-regime mode: no-go-theorem
- Bridge object: unspecified witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'unspecified witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'. It must preserve the invariant 'consequence schema NP-complete-classification'. Its operational role is 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.'.
- First nontrivial lemma: If 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.' fails for 'unspecified witness family', then no route preserving 'consequence schema NP-complete-classification' can force 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' from 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'.
- Quantifier signature: For every admissible 'unspecified witness family' on the route from 'st.thm.karp-many-natural-problems-are-np-complete-karp1' to 'st.thm.murray-williams-non-np-hardness-of-mcsp-under-lo'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Every attempted realization of 'unspecified witness family' remains blocked by the current barrier, so 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' does not follow.
- Proof strategy family: barrier
- Negative route: Assume every attempted realization of 'unspecified witness family' collapses into the current barrier.
- Failure diagnosis: The route fails if every admissible realization of 'unspecified witness family' remains trapped behind the current barrier.
- Proof obligations: If 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.' fails for 'unspecified witness family', then no route preserving 'consequence schema NP-complete-classification' can force 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' from 'There exist many problems L in combinatorial

optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'.

- Candidate lemmas: Prove that a hardness-magnification witness family turns 'NP-complete under polynomial-time many-one reductions' into 'is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various...' while preserving 'consequence schema NP-complete-classification'.; Barrier obstruction lemma for unspecified witness family
- Independent value: A no-go theorem for 'unspecified witness family' would be independently valuable because it collapses an entire discovery subspace and exposes a genuine barrier theorem.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.02

- Exact statement: There exists an admissible 'unspecified reduction gadget family' such that 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.' can be organized into a route preserving 'consequence schema NP-complete-classification' and thereby supports 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.'.
- Restricted-regime mode: no-go-theorem
- Bridge object: unspecified reduction gadget family (family of reductions preserving the route witness structure)
- Bridge definition: 'unspecified reduction gadget family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'. It must preserve the invariant 'consequence schema NP-complete-classification'. Its operational role is 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.'.
- First nontrivial lemma: If 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.' fails for 'unspecified reduction gadget family', then no route preserving 'consequence schema NP-complete-classification' can force 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' from 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'.
- Quantifier signature: For every admissible 'unspecified reduction gadget family' on the route from 'st.thm.karp-many-natural-problems-are-np-complete-karp1' to 'st.thm.murray-williams-non-np-hardness-of-mcsp-under-lo'.

- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Every attempted realization of 'unspecified reduction gadget family' remains blocked by the current barrier, so 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' does not follow.
- Proof strategy family: barrier
- Negative route: Assume every attempted realization of 'unspecified reduction gadget family' collapses into the current barrier.
- Failure diagnosis: The route fails if every admissible realization of 'unspecified reduction gadget family' remains trapped behind the current barrier.
- Proof obligations: If 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.' fails for 'unspecified reduction gadget family', then no route preserving 'consequence schema NP-complete-classification' can force 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' from 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'.
- Candidate lemmas: Prove that a hardness-magnification witness family turns 'NP-complete under polynomial-time many-one reductions' into 'is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various...' while preserving 'consequence schema NP-complete-classification'.; Barrier obstruction lemma for unspecified reduction gadget family
- Independent value: A no-go theorem for 'unspecified reduction gadget family' would be independently valuable because it collapses an entire discovery subspace and exposes a genuine barrier theorem.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.03

- Exact statement: No admissible 'unspecified witness family' can simultaneously preserve 'consequence schema NP-complete-classification' and realize 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' under the current route commitments.
- Restricted-regime mode: no-go-theorem
- Bridge object: unspecified witness family (family of instances equipped with a monotone witness measure)

- Bridge definition: 'unspecified witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'. It must preserve the invariant 'consequence schema NP-complete-classification'. Its operational role is 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.'.
- First nontrivial lemma: If 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.' fails for 'unspecified witness family', then no route preserving 'consequence schema NP-complete-classification' can force 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' from 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'.
- Quantifier signature: For every admissible 'unspecified witness family' on the route from 'st.thm.karp-many-natural-problems-are-np-complete-karp1' to 'st.thm.murray-williams-non-np-hardness-of-mcsp-under-lo'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Every attempted realization of 'unspecified witness family' remains blocked by the current barrier, so 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' does not follow.
- Proof strategy family: barrier
- Negative route: Assume every attempted realization of 'unspecified witness family' collapses into the current barrier.
- Failure diagnosis: The route fails if every admissible realization of 'unspecified witness family' remains trapped behind the current barrier.
- Proof obligations: If 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.' fails for 'unspecified witness family', then no route preserving 'consequence schema NP-complete-classification' can force 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' from 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'.
- Candidate lemmas: Obstruction theorem for unspecified witness family; Barrier obstruction lemma for unspecified witness family

- Independent value: A no-go theorem for 'unspecified witness family' would be independently valuable because it collapses an entire discovery subspace and exposes a genuine barrier theorem.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.04

- Exact statement: There exists an admissible 'unspecified witness family' whose realization internalizes 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.' while preserving 'consequence schema NP-complete-classification' strongly enough to support 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.'.
- Restricted-regime mode: no-go-theorem
- Bridge object: unspecified witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'unspecified witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'. It must preserve the invariant 'consequence schema NP-complete-classification'. Its operational role is 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.'.
- First nontrivial lemma: If 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.' fails for 'unspecified witness family', then no route preserving 'consequence schema NP-complete-classification' can force 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' from 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'.
- Quantifier signature: For every admissible 'unspecified witness family' on the route from 'st.thm.karp-many-natural-problems-are-np-complete-karp1' to 'st.thm.murray-williams-non-np-hardness-of-mcsp-under-lo'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Every attempted realization of 'unspecified witness family' remains blocked by the current barrier, so 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' does not follow.

- Proof strategy family: barrier
- Negative route: Assume every attempted realization of 'unspecified witness family' collapses into the current barrier.
- Failure diagnosis: The route fails if every admissible realization of 'unspecified witness family' remains trapped behind the current barrier.
- Proof obligations: If 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.' fails for 'unspecified witness family', then no route preserving 'consequence schema NP-complete-classification' can force 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' from 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'.
- Candidate lemmas: Realization theorem for unspecified witness family; Barrier obstruction lemma for unspecified witness family
- Independent value: A no-go theorem for 'unspecified witness family' would be independently valuable because it collapses an entire discovery subspace and exposes a genuine barrier theorem.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.05

- Exact statement: No admissible 'unspecified reduction gadget family' can simultaneously preserve 'consequence schema NP-complete-classification' and realize 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' under the current route commitments.
- Restricted-regime mode: no-go-theorem
- Bridge object: unspecified reduction gadget family (family of reductions preserving the route witness structure)
- Bridge definition: 'unspecified reduction gadget family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'. It must preserve the invariant 'consequence schema NP-complete-classification'. Its operational role is 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.'.
- First nontrivial lemma: If 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.' fails for 'unspecified reduction gadget family', then no route preserving 'consequence schema NP-complete-classification' can force 'Under"local"polynomial-time reductions (where each output bit is

computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' from 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'.

- Quantifier signature: For every admissible 'unspecified reduction gadget family' on the route from 'st.thm.karp-many-natural-problems-are-np-complete-karp1' to 'st.thm.murray-williams-non-np-hardness-of-mcsp-under-lo'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Every attempted realization of 'unspecified reduction gadget family' remains blocked by the current barrier, so 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' does not follow.
- Proof strategy family: barrier
- Negative route: Assume every attempted realization of 'unspecified reduction gadget family' collapses into the current barrier.
- Failure diagnosis: The route fails if every admissible realization of 'unspecified reduction gadget family' remains trapped behind the current barrier.
- Proof obligations: If 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.' fails for 'unspecified reduction gadget family', then no route preserving 'consequence schema NP-complete-classification' can force 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' from 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'.
- Candidate lemmas: Obstruction theorem for unspecified reduction gadget family; Barrier obstruction lemma for unspecified reduction gadget family
- Independent value: A no-go theorem for 'unspecified reduction gadget family' would be independently valuable because it collapses an entire discovery subspace and exposes a genuine barrier theorem.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-barrier-aware-transfer.06

- Exact statement: There exists an admissible 'unspecified reduction gadget family' whose realization internalizes 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.' while preserving

'consequence schema NP-complete-classification' strongly enough to support 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.'.

- Restricted-regime mode: no-go-theorem
- Bridge object: unspecified reduction gadget family (family of reductions preserving the route witness structure)
- Bridge definition: 'unspecified reduction gadget family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'. It must preserve the invariant 'consequence schema NP-complete-classification'. Its operational role is 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.'.
- First nontrivial lemma: If 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.' fails for 'unspecified reduction gadget family', then no route preserving 'consequence schema NP-complete-classification' can force 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' from 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'.
- Quantifier signature: For every admissible 'unspecified reduction gadget family' on the route from 'st.thm.karp-many-natural-problems-are-np-complete-karp1' to 'st.thm.murray-williams-non-np-hardness-of-mcsp-under-lo'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Every attempted realization of 'unspecified reduction gadget family' remains blocked by the current barrier, so 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' does not follow.
- Proof strategy family: barrier
- Negative route: Assume every attempted realization of 'unspecified reduction gadget family' collapses into the current barrier.
- Failure diagnosis: The route fails if every admissible realization of 'unspecified reduction gadget family' remains trapped behind the current barrier.

- Proof obligations: If 'Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.' fails for 'unspecified reduction gadget family', then no route preserving 'consequence schema NP-complete-classification' can force 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.' from 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.'.
- Candidate lemmas: Realization theorem for unspecified reduction gadget family; Barrier obstruction lemma for unspecified reduction gadget family
- Independent value: A no-go theorem for 'unspecified reduction gadget family' would be independently valuable because it collapses an entire discovery subspace and exposes a genuine barrier theorem.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.01

- Exact statement: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP

slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Prove that a hardness-magnification witness family turns 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' into 'major consequences such as NP not in P/poly' while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.02

- Exact statement: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models

(e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.

- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Candidate lemmas: Prove that a hardness-magnification witness family preserves 'MCSP complexity profile in nonuniform-circuits' while transporting the route from the nonuniform-circuits regime to the formula-size regime strongly enough to derive 'major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter...'.
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.03

- Exact statement: There exists an admissible 'MCSP obstruction family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: special-case
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)
- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.'.

- First nontrivial lemma: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Prove that a hardness-magnification witness family turns 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' into 'major consequences such as NP not in P/poly' while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Independent value: A concrete special-case theorem for 'MCSP obstruction family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.04

- Exact statement: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.

- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Obstruction theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.05

- Exact statement: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Realization theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.06

- Exact statement: There exists an intermediate typed object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through theorem-sized maps preserving 'MCSP complexity profile in nonuniform-circuits'.
- Restricted-regime mode: special-case

- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- Candidate lemmas: Factorization theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.07

- Exact statement: For every admissible 'MCSP witness family', there is a transport law from 'nonuniform-circuits' to 'formula-size' that preserves 'MCSP complexity profile in nonuniform-circuits' and retains enough strength to activate 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).

- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Candidate lemmas: Transport theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.08

- Exact statement: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' under the current route commitments.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model

implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.'.

- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Candidate lemmas: Obstruction theorem for MCSP witness family

- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.09

- Exact statement: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).

- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Candidate lemmas: Realization theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.10

- Exact statement: There exists an intermediate typed object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' factors through theorem-sized maps preserving 'MCSP complexity profile in nonuniform-circuits'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)

- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models

(including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- Candidate lemmas: Factorization theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.11

- Exact statement: No admissible 'MCSP obstruction family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.
- Restricted-regime mode: special-case
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)
- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.'.
- First nontrivial lemma: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).

- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Obstruction theorem for MCSP obstruction family
- Independent value: A concrete special-case theorem for 'MCSP obstruction family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.12

- Exact statement: There exists an admissible 'MCSP obstruction family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: special-case
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)
- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is

a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.'.

- First nontrivial lemma: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Realization theorem for MCSP obstruction family
- Independent value: A concrete special-case theorem for 'MCSP obstruction family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-hybridization-of-partial-bridges.13

- Exact statement: There exists an intermediate typed object built from 'MCSP obstruction family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through theorem-sized maps preserving 'MCSP complexity profile in nonuniform-circuits'.
- Restricted-regime mode: special-case
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)
- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.'.
- First nontrivial lemma: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: constructive-realization

- Negative route: Assume 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Factorization theorem for MCSP obstruction family
- Independent value: A concrete special-case theorem for 'MCSP obstruction family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-regime-localization.01

- Exact statement: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: threshold-version
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'The frontier repeatedly crosses model or threshold boundaries, so discovery should search for the smallest regime in which the route first becomes true.'.
- First nontrivial lemma: There exists a threshold regime in which 'MCSP witness family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a

restricted version of 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}exists \{}theta,\{} \{}forall W,\{} \{}mathrm{Adm}_{\{}theta}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarg
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: threshold-localization
- Negative route: Assume no threshold-localized version of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough for the restricted target.
- Failure diagnosis: The route fails if no threshold-localized version of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough to force the restricted target.
- Proof obligations: There exists a threshold regime in which 'MCSP witness family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Prove that a hardness-magnification witness family turns 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' into 'major consequences such as NP not in P/poly' while preserving 'MCSP complexity profile in nonuniform-circuits'.; Threshold localization lemma for MCSP witness family
- Independent value: A threshold theorem for 'MCSP witness family' would be independently valuable because it localizes the first nontrivial regime where the route becomes mathematically real.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-regime-localization.02

- Exact statement: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models

(e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- Restricted-regime mode: threshold-version
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'The frontier repeatedly crosses model or threshold boundaries, so discovery should search for the smallest regime in which the route first becomes true.'.
- First nontrivial lemma: There exists a threshold regime in which 'MCSP witness family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}exists \{}theta,\{} \{}forall W,\{} \{}mathrm{Adm}_{\{}theta}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarg
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Proof strategy family: threshold-localization
- Negative route: Assume no threshold-localized version of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough for the restricted target.
- Failure diagnosis: The route fails if no threshold-localized version of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough to force the restricted target.
- Proof obligations: There exists a threshold regime in which 'MCSP witness family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'There exist explicit"meta"problems (including MCSP variants) such that proving

certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- Candidate lemmas: Prove that a hardness-magnification witness family preserves 'MCSP complexity profile in nonuniform-circuits' while transporting the route from the nonuniform-circuits regime to the formula-size regime strongly enough to derive 'major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter...'.; Threshold localization lemma for MCSP witness family
- Independent value: A threshold theorem for 'MCSP witness family' would be independently valuable because it localizes the first nontrivial regime where the route becomes mathematically real.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-regime-localization.03

- Exact statement: There exists an admissible 'MCSP obstruction family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: threshold-version
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)
- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'The frontier repeatedly crosses model or threshold boundaries, so discovery should search for the smallest regime in which the route first becomes true.'.
- First nontrivial lemma: There exists a threshold regime in which 'MCSP obstruction family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.

- LaTeX theorem: \{}exists \{}theta,\{} \{}forall W,\{} \{}mathrm{Adm}_{\{}theta}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarg
- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: threshold-localization
- Negative route: Assume no threshold-localized version of 'MCSP obstruction family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough for the restricted target.
- Failure diagnosis: The route fails if no threshold-localized version of 'MCSP obstruction family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough to force the restricted target.
- Proof obligations: There exists a threshold regime in which 'MCSP obstruction family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Prove that a hardness-magnification witness family turns 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' into 'major consequences such as NP not in P/poly' while preserving 'MCSP complexity profile in nonuniform-circuits'.; Threshold localization lemma for MCSP obstruction family
- Independent value: A threshold theorem for 'MCSP obstruction family' would be independently valuable because it localizes the first nontrivial regime where the route becomes mathematically real.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-regime-localization.04

- Exact statement: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.
- Restricted-regime mode: threshold-version
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem

(e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'The frontier repeatedly crosses model or threshold boundaries, so discovery should search for the smallest regime in which the route first becomes true.'.

- First nontrivial lemma: There exists a threshold regime in which 'MCSP witness family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: threshold-localization
- Negative route: Assume no threshold-localized version of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough for the restricted target.
- Failure diagnosis: The route fails if no threshold-localized version of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough to force the restricted target.
- Proof obligations: There exists a threshold regime in which 'MCSP witness family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Obstruction theorem for MCSP witness family; Threshold localization lemma for MCSP witness family
- Independent value: A threshold theorem for 'MCSP witness family' would be independently valuable because it localizes the first nontrivial regime where the route becomes mathematically real.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-regime-localization.05

- Exact statement: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for

an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- Restricted-regime mode: threshold-version
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'The frontier repeatedly crosses model or threshold boundaries, so discovery should search for the smallest regime in which the route first becomes true.'.
- First nontrivial lemma: There exists a threshold regime in which 'MCSP witness family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: threshold-localization
- Negative route: Assume no threshold-localized version of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough for the restricted target.
- Failure diagnosis: The route fails if no threshold-localized version of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough to force the restricted target.
- Proof obligations: There exists a threshold regime in which 'MCSP witness family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers

(e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- Candidate lemmas: Realization theorem for MCSP witness family; Threshold localization lemma for MCSP witness family
- Independent value: A threshold theorem for 'MCSP witness family' would be independently valuable because it localizes the first nontrivial regime where the route becomes mathematically real.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-regime-localization.06

- Exact statement: For every admissible 'MCSP witness family', there is a transport law from 'nonuniform-circuits' to 'formula-size' that preserves 'MCSP complexity profile in nonuniform-circuits' and retains enough strength to activate 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Restricted-regime mode: threshold-version
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'The frontier repeatedly crosses model or threshold boundaries, so discovery should search for the smallest regime in which the route first becomes true.'.
- First nontrivial lemma: There exists a threshold regime in which 'MCSP witness family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).

- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Proof strategy family: threshold-localization
- Negative route: Assume no threshold-localized version of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough for the restricted target.
- Failure diagnosis: The route fails if no threshold-localized version of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough to force the restricted target.
- Proof obligations: There exists a threshold regime in which 'MCSP witness family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Candidate lemmas: Transport theorem for MCSP witness family; Threshold localization lemma for MCSP witness family
- Independent value: A threshold theorem for 'MCSP witness family' would be independently valuable because it localizes the first nontrivial regime where the route becomes mathematically real.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-regime-localization.07

- Exact statement: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' under the current route commitments.
- Restricted-regime mode: threshold-version
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem

(e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'The frontier repeatedly crosses model or threshold boundaries, so discovery should search for the smallest regime in which the route first becomes true.'.

- First nontrivial lemma: There exists a threshold regime in which 'MCSP witness family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Proof strategy family: threshold-localization
- Negative route: Assume no threshold-localized version of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough for the restricted target.
- Failure diagnosis: The route fails if no threshold-localized version of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough to force the restricted target.
- Proof obligations: There exists a threshold regime in which 'MCSP witness family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Candidate lemmas: Obstruction theorem for MCSP witness family; Threshold localization lemma for MCSP witness family
- Independent value: A threshold theorem for 'MCSP witness family' would be independently valuable because it localizes the first nontrivial regime where the route becomes mathematically real.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-regime-localization.08

- Exact statement: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Restricted-regime mode: threshold-version
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'The frontier repeatedly crosses model or threshold boundaries, so discovery should search for the smallest regime in which the route first becomes true.'.
- First nontrivial lemma: There exists a threshold regime in which 'MCSP witness family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Proof strategy family: threshold-localization

- Negative route: Assume no threshold-localized version of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough for the restricted target.
- Failure diagnosis: The route fails if no threshold-localized version of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough to force the restricted target.
- Proof obligations: There exists a threshold regime in which 'MCSP witness family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Candidate lemmas: Realization theorem for MCSP witness family; Threshold localization lemma for MCSP witness family
- Independent value: A threshold theorem for 'MCSP witness family' would be independently valuable because it localizes the first nontrivial regime where the route becomes mathematically real.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-regime-localization.09

- Exact statement: No admissible 'MCSP obstruction family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.
- Restricted-regime mode: threshold-version
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)
- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'The frontier repeatedly crosses model or threshold boundaries, so discovery should search for the smallest regime in which the route first becomes true.'.
- First nontrivial lemma: There exists a threshold regime in which 'MCSP obstruction family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: threshold-localization
- Negative route: Assume no threshold-localized version of 'MCSP obstruction family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough for the restricted target.
- Failure diagnosis: The route fails if no threshold-localized version of 'MCSP obstruction family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough to force the restricted target.
- Proof obligations: There exists a threshold regime in which 'MCSP obstruction family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Obstruction theorem for MCSP obstruction family; Threshold localization lemma for MCSP obstruction family
- Independent value: A threshold theorem for 'MCSP obstruction family' would be independently valuable because it localizes the first nontrivial regime where the route becomes mathematically real.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-regime-localization.10

- Exact statement: There exists an admissible 'MCSP obstruction family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: threshold-version
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)

- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'The frontier repeatedly crosses model or threshold boundaries, so discovery should search for the smallest regime in which the route first becomes true.'.
- First nontrivial lemma: There exists a threshold regime in which 'MCSP obstruction family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: threshold-localization
- Negative route: Assume no threshold-localized version of 'MCSP obstruction family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough for the restricted target.
- Failure diagnosis: The route fails if no threshold-localized version of 'MCSP obstruction family' preserves 'MCSP complexity profile in nonuniform-circuits' strongly enough to force the restricted target.
- Proof obligations: There exists a threshold regime in which 'MCSP obstruction family' already preserves 'MCSP complexity profile in nonuniform-circuits' well enough to imply a restricted version of 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Realization theorem for MCSP obstruction family; Threshold localization lemma for MCSP obstruction family
- Independent value: A threshold theorem for 'MCSP obstruction family' would be independently valuable because it localizes the first nontrivial regime where the route becomes mathematically real.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-abstraction-search.01

- Exact statement: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Find a sharper abstraction that turns the blocked route into a typed, inspectable theorem target.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.

- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Prove that a hardness-magnification witness family turns 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' into 'major consequences such as NP not in P/poly' while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-abstraction-search.02

- Exact statement: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Find a sharper abstraction that turns the blocked route into a typed, inspectable theorem target.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the

passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Candidate lemmas: Prove that a hardness-magnification witness family preserves 'MCSP complexity profile in nonuniform-circuits' while transporting the route from the nonuniform-circuits regime to the formula-size regime strongly enough to derive 'major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter...'.
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.

- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-abstraction-search.03

- Exact statement: There exists an admissible 'MCSP obstruction family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: special-case
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)
- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Find a sharper abstraction that turns the blocked route into a typed, inspectable theorem target.'.
- First nontrivial lemma: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: constructive-realization

- Negative route: Assume 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Prove that a hardness-magnification witness family turns 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' into 'major consequences such as NP not in P/poly' while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Independent value: A concrete special-case theorem for 'MCSP obstruction family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-abstraction-search.04

- Exact statement: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Find a sharper abstraction that turns the blocked route into a typed, inspectable theorem target.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower

bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Obstruction theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-abstraction-search.05

- Exact statement: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly

beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Find a sharper abstraction that turns the blocked route into a typed, inspectable theorem target.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong

model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- Candidate lemmas: Realization theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-abstraction-search.06

- Exact statement: For every admissible 'MCSP witness family', there is a transport law from 'nonuniform-circuits' to 'formula-size' that preserves 'MCSP complexity profile in nonuniform-circuits' and retains enough strength to activate 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Find a sharper abstraction that turns the blocked route into a typed, inspectable theorem target.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.

- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Candidate lemmas: Transport theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-abstraction-search.07

- Exact statement: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' under the current route commitments.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)

- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Find a sharper abstraction that turns the blocked route into a typed, inspectable theorem target.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models

(including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- Candidate lemmas: Obstruction theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-abstraction-search.08

- Exact statement: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Find a sharper abstraction that turns the blocked route into a typed, inspectable theorem target.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.

- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Candidate lemmas: Realization theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-abstraction-search.09

- Exact statement: No admissible 'MCSP obstruction family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.
- Restricted-regime mode: special-case
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)

- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Find a sharper abstraction that turns the blocked route into a typed, inspectable theorem target.'.
- First nontrivial lemma: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Obstruction theorem for MCSP obstruction family

- Independent value: A concrete special-case theorem for 'MCSP obstruction family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-abstraction-search.10

- Exact statement: There exists an admissible 'MCSP obstruction family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: special-case
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)
- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Find a sharper abstraction that turns the blocked route into a typed, inspectable theorem target.'.
- First nontrivial lemma: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Realization theorem for MCSP obstruction family
- Independent value: A concrete special-case theorem for 'MCSP obstruction family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.01

- Exact statement: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Several live routes are still searching for the right preserved quantity or witness measure, so invariant discovery remains a first-class discovery move.'.

- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: invariant-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Prove that a hardness-magnification witness family turns 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' into 'major consequences such as NP not in P/poly' while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.02

- Exact statement: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Several live routes are still searching for the right preserved quantity or witness measure, so invariant discovery remains a first-class discovery move.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- Proof strategy family: invariant-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Candidate lemmas: Prove that a hardness-magnification witness family preserves 'MCSP complexity profile in nonuniform-circuits' while transporting the route from the nonuniform-circuits regime to the formula-size regime strongly enough to derive 'major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter...'.
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.03

- Exact statement: There exists an admissible 'MCSP obstruction family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: special-case
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)
- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a

meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Several live routes are still searching for the right preserved quantity or witness measure, so invariant discovery remains a first-class discovery move.'.

- First nontrivial lemma: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: invariant-realization
- Negative route: Assume 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Prove that a hardness-magnification witness family turns 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' into 'major consequences such as NP not in P/poly' while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Independent value: A concrete special-case theorem for 'MCSP obstruction family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.

- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.04

- Exact statement: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Several live routes are still searching for the right preserved quantity or witness measure, so invariant discovery remains a first-class discovery move.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: invariant-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.

- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Obstruction theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.05

- Exact statement: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Several live routes are still searching for the right preserved quantity or witness measure, so invariant discovery remains a first-class discovery move.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP

slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: invariant-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Realization theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.06

- Exact statement: For every admissible 'MCSP witness family', there is a transport law from 'nonuniform-circuits' to 'formula-size' that preserves 'MCSP complexity profile in nonuniform-circuits' and retains enough strength to activate 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Restricted-regime mode: special-case

- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Several live routes are still searching for the right preserved quantity or witness measure, so invariant discovery remains a first-class discovery move.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Proof strategy family: invariant-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP

variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- Candidate lemmas: Transport theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.07

- Exact statement: No admissible 'MCSP witness family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' under the current route commitments.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Several live routes are still searching for the right preserved quantity or witness measure, so invariant discovery remains a first-class discovery move.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.

- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Proof strategy family: invariant-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Candidate lemmas: Obstruction theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.08

- Exact statement: There exists an admissible 'MCSP witness family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Restricted-regime mode: special-case

- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Several live routes are still searching for the right preserved quantity or witness measure, so invariant discovery remains a first-class discovery move.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly)'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly)'.
- Proof strategy family: invariant-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP

variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- Candidate lemmas: Realization theorem for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.09

- Exact statement: No admissible 'MCSP obstruction family' can simultaneously preserve 'MCSP complexity profile in nonuniform-circuits' and realize 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' under the current route commitments.
- Restricted-regime mode: special-case
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)
- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Several live routes are still searching for the right preserved quantity or witness measure, so invariant discovery remains a first-class discovery move.'.
- First nontrivial lemma: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}neg \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).

- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: invariant-realization
- Negative route: Assume 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Obstruction theorem for MCSP obstruction family
- Independent value: A concrete special-case theorem for 'MCSP obstruction family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-pattern-invariant-discovery.10

- Exact statement: There exists an admissible 'MCSP obstruction family' whose realization internalizes 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to support 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Restricted-regime mode: special-case
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)
- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a

meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Several live routes are still searching for the right preserved quantity or witness measure, so invariant discovery remains a first-class discovery move.'.

- First nontrivial lemma: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Proof strategy family: invariant-realization
- Negative route: Assume 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Proof obligations: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Candidate lemmas: Realization theorem for MCSP obstruction family
- Independent value: A concrete special-case theorem for 'MCSP obstruction family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.01

- Exact statement: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.', and 'MCSP complexity profile in nonuniform-circuits' transports coherently along that witness.
- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' is not licensed. no_equivalence_witness
- Proof strategy family: equivalence-transport
- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.
- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'MCSP complexity profile in nonuniform-circuits' to the target-side operationalization. Local negative probe: no_equivalence_witness.

- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.
- Candidate lemmas: Equivalence and coherent transport theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family
- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.02

- Exact statement: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).', and 'MCSP complexity profile in nonuniform-circuits' transports coherently along that witness.
- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.

- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).' is not licensed. no_equivalence_witness
- Proof strategy family: equivalence-transport
- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.
- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'MCSP complexity profile in nonuniform-circuits' to the target-side operationalization. Local negative probe: no_equivalence_witness.
- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'MCSP complexity profile in nonuniform-circuits' toward 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.; Transport obligation: move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.
- Candidate lemmas: Equivalence and coherent transport theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family
- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-equivalence-coherence-reframing.03

- Exact statement: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification

thresholds below previously known lower bounds in some settings.', and 'exponential-lower-bound witness complexity' transports coherently along that witness.

- Restricted-regime mode: equivalence-theorem
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'exponential-lower-bound witness complexity'. Its operational role is 'Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'exponential-lower-bound witness complexity' toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.atserias-m-uller-simple-general-magnification-fr'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: Absent an explicit equivalence witness for 'MCSP witness family', transport toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.' is not licensed. no_equivalence_witness
- Proof strategy family: equivalence-transport
- Negative route: Assume no equivalence witness for 'MCSP witness family' aligns the route-side and target-side objects.
- Failure diagnosis: The route fails if no equivalence witness for 'MCSP witness family' transports 'exponential-lower-bound witness complexity' to the target-side operationalization. Local negative probe: no_equivalence_witness.
- Proof obligations: For every admissible 'MCSP witness family', the target-side operationalization is equivalent to the route-side operationalization in the sense needed to transport 'exponential-lower-bound witness complexity' toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for distinguishers/sparse-code objects can be amplified into stronger circuit/formula lower bounds, achieving magnification thresholds below previously known lower bounds in some settings.'.; Transport obligation:

move the invariant across the proposed equivalence for 'carrier-realization' without changing the target claim.; Transport obligation: move the invariant across the proposed equivalence for 'formula-size-transport' without changing the target claim.; Equivalence claim: determine whether the live carrier for 'carrier-realization' is identical to, equivalent to, or only analogous with the target-side object class.

- Candidate lemmas: Equivalence and coherent transport theorem for MCSP witness family; Equivalence transport lemma for MCSP witness family
- Independent value: An equivalence theorem for 'MCSP witness family' would independently matter because it identifies when two live operationalizations are mathematically interchangeable rather than merely analogous.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-categorical-reframing.01

- Exact statement: There exists a mediating object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute.
- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'. non_commuting_route
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.
- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.
- Candidate lemmas: Commutative factorization theorem for MCSP witness family; Coherence diagram lemma for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-categorical-reframing.02

- Exact statement: There exists a mediating object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter

regimes consequences as strong as NP not in P/poly).' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute.

- Restricted-regime mode: special-case
- Bridge object: MCSP witness family (family of instances equipped with a monotone witness measure)
- Bridge definition: 'MCSP witness family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.
- First nontrivial lemma: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.
- Quantifier signature: For every admissible 'MCSP witness family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP witness family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'. non_commuting_route
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP witness family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.

- Proof obligations: For every admissible 'MCSP witness family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'There exist explicit"meta"problems (including MCSP variants) such that proving certain emph barely-superlinear lower bounds against restricted models (e.g. formulas) would imply major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter regimes consequences as strong as NP not in P/poly).'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.
- Candidate lemmas: Commutative factorization theorem for MCSP witness family; Coherence diagram lemma for MCSP witness family
- Independent value: A concrete special-case theorem for 'MCSP witness family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6

conjpkg.disclayer-p-vs-np-family-categorical-reframing.03

- Exact statement: There exists a mediating object built from 'MCSP obstruction family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute.
- Restricted-regime mode: special-case
- Bridge object: MCSP obstruction family (family of hard instances or obstructions forcing the target payoff)
- Bridge definition: 'MCSP obstruction family' denotes the structured intermediate object required by this branch. It is meant to organize the source-side content 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.'. It must preserve the invariant 'MCSP complexity profile in nonuniform-circuits'. Its operational role is 'Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.'.

- First nontrivial lemma: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Quantifier signature: For every admissible 'MCSP obstruction family' on the route from 'st.def.hardness-magnification-phenomenon-programme-leve' to 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- LaTeX theorem: \{}forall W,\{} \{}mathrm{Adm}(W) \{}to \{}exists w : \{}mathsf{BW}(W),\{} \{}mathrm{PresInv}(W,w) \{}wedge \{}mathrm{RealizesTarget}(W,w).
- Negative/barrier version: If 'MCSP obstruction family' cannot be realized with invariant 'MCSP complexity profile in nonuniform-circuits', the route cannot force 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'. non_commuting_route
- Proof strategy family: constructive-realization
- Negative route: Assume 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits' strongly enough to activate the local target claim.
- Failure diagnosis: The route fails if 'MCSP obstruction family' cannot be realized while preserving 'MCSP complexity profile in nonuniform-circuits'. Local negative probe: non_commuting_route.
- Proof obligations: For every admissible 'MCSP obstruction family', the first route-specific realization step preserves 'MCSP complexity profile in nonuniform-circuits' and prepares the passage from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.; Commutativity obligation: the route for 'carrier-realization' must commute between source theorem hypotheses, the mediating carrier, and the target theorem hypotheses.
- Candidate lemmas: Commutative factorization theorem for MCSP obstruction family; Coherence diagram lemma for MCSP obstruction family
- Independent value: A concrete special-case theorem for 'MCSP obstruction family' would be independently valuable because it converts a broad discovery programme into an inspectable mathematical subresult.
- Expert gate: 6/6